Homework 3

Part I: Chemical Reaction

The methane oxidation reaction is

$$CH_4 + 2O_2 \rightarrow 2H_2O + CO_2. \tag{1}$$

Given, the temperature is 800 K and the total gas concentration is

$$[C] = 0.02 \text{ kmol/m}^3 = 20 \text{ mol/m}^3. \tag{2}$$

Gibbs Free Energy Calculation

The Gibbs free energy of each species is

$$\bar{g}_i = \bar{g}_i^\circ + RT \ln X_i. \tag{3}$$

Therefore, the Gibbs free energy change for the methane oxidation reaction is

$$\Delta\bar{g}_{rxn} = (2\bar{g}_{H_2O} + \bar{g}_{CO_2}) - (\bar{g}_{CH_4} + 2\bar{g}_{O_2}). \tag{4}$$

Now,

$$\Delta\bar{g}_{rxn}^\circ = [2(-402.89) + (-576.71)] - [(-236.14) + 2(-172.93)]. \tag{5}$$

$$\Delta\bar{g}_{rxn}^\circ = -1382.49 - (-582.00) = -800.49 \text{ kJ/mol}. \tag{6}$$

The composition species effect included using the reaction quotient,

$$Q = \frac{X_{H_2O}^2 X_{CO_2}}{X_{CH_4} X_{O_2}^2}. \tag{7}$$

Since

$$R = 0.008314 \text{ kJ/(mol} \cdot \text{K)} \tag{8}$$

and

$$RT = (0.008314)(800) = 6.6512 \text{ kJ/mol}, \tag{9}$$

hence the equation used for each case is

$$\Delta\bar{g}_{rxn} = -800.49 + 6.6512 \ln\left(\frac{X_{H_2O}^2 X_{CO_2}}{X_{CH_4} X_{O_2}^2}\right). \tag{10}$$

Reaction Rate Calculation

The rate of progress is calculated using mass-action kinetics:

$$q_{rxn} = k_{fwd} C_{CH_4} C_{O_2}^2 - k_{rev} C_{CO_2} C_{H_2O}^2. \tag{11}$$

Here,

$$k_{fwd} = 2.10 \times 10^{13}\ \mathrm{m}^6/(\mathrm{mol}^2 \cdot \mathrm{s}) \tag{12}$$

and

$$k_{rev} = 1.81 \times 10^{10}\ \mathrm{m}^6/(\mathrm{mol}^2 \cdot \mathrm{s}). \tag{13}$$

For each species, the concentration is obtained from its mole fraction:

$$C_i = X_i C_{total}. \tag{14}$$

Since $C_{total} = 20\ \mathrm{mol/m^3}$, each mole fraction is multiplied by 20.

Case 1

For Case 1,

$$X_{CH_4} = 0.147, \qquad X_{CO_2} = 0.069, \qquad X_{H_2O} = 0.460, \qquad X_{O_2} = 0.324. \tag{15}$$

The reaction quotient is

$$Q_1 = \frac{(0.460)^2(0.069)}{(0.147)(0.324)^2} = 0.9461. \tag{16}$$

Therefore,

$$\Delta\bar{g}_{rxn,1} = -800.49 + 6.6512 \ln(0.9461) = -800.86\ \mathrm{kJ/mol}. \tag{17}$$

The concentrations are

$$C_{CH_4} = 0.147(20) = 2.94\ \mathrm{mol/m^3}, \tag{18}$$
$$C_{O_2} = 0.324(20) = 6.48\ \mathrm{mol/m^3}, \tag{19}$$
$$C_{CO_2} = 0.069(20) = 1.38\ \mathrm{mol/m^3}, \tag{20}$$
$$C_{H_2O} = 0.460(20) = 9.20\ \mathrm{mol/m^3}. \tag{21}$$

Substituting these into the rate expression,

$$q_{rxn,1} = (2.10 \times 10^{13})(2.94)(6.48)^2 - (1.81 \times 10^{10})(1.38)(9.20)^2. \tag{22}$$

$$q_{rxn,1} = 2.590 \times 10^{15} \text{ mol/(m}^3 \cdot \text{s)}. \tag{23}$$

Case 2

For Case 2,

$$X_{CH_4} = 0.147, \qquad X_{CO_2} = 0.069, \qquad X_{H_2O} = 0.579, \qquad X_{O_2} = 0.206. \tag{24}$$

$$Q_2 = \frac{(0.579)^2(0.069)}{(0.147)(0.206)^2} = 3.7081. \tag{25}$$

$$\Delta\bar{g}_{rxn,2} = -800.49 + 6.6512 \ln(3.7081) = -791.77 \text{ kJ/mol}. \tag{26}$$

The concentrations are

$$C_{CH_4} = 0.147(20) = 2.94 \text{ mol/m}^3, \tag{27}$$
$$C_{O_2} = 0.206(20) = 4.12 \text{ mol/m}^3, \tag{28}$$
$$C_{CO_2} = 0.069(20) = 1.38 \text{ mol/m}^3, \tag{29}$$
$$C_{H_2O} = 0.579(20) = 11.58 \text{ mol/m}^3. \tag{30}$$

$$q_{rxn,2} = (2.10 \times 10^{13})(2.94)(4.12)^2 - (1.81 \times 10^{10})(1.38)(11.58)^2. \tag{31}$$

$$q_{rxn,2} = 1.045 \times 10^{15} \text{ mol/(m}^3 \cdot \text{s)}. \tag{32}$$

Case 3

For Case 3,

$$X_{CH_4} = 0.005, \qquad X_{CO_2} = 0.339, \qquad X_{H_2O} = 0.396, \qquad X_{O_2} = 0.260. \tag{33}$$

$$Q_3 = \frac{(0.396)^2(0.339)}{(0.005)(0.260)^2} = 157.28. \tag{34}$$

$$\Delta\bar{g}_{rxn,3} = -800.49 + 6.6512 \ln(157.28) = -766.85 \text{ kJ/mol}. \tag{35}$$

The concentrations are

$$C_{CH_4} = 0.005(20) = 0.10 \text{ mol/m}^3, \tag{36}$$
$$C_{O_2} = 0.260(20) = 5.20 \text{ mol/m}^3, \tag{37}$$
$$C_{CO_2} = 0.339(20) = 6.78 \text{ mol/m}^3, \tag{38}$$
$$C_{H_2O} = 0.396(20) = 7.92 \text{ mol/m}^3. \tag{39}$$

$$q_{rxn,3} = (2.10 \times 10^{13})(0.10)(5.20)^2 - (1.81 \times 10^{10})(6.78)(7.92)^2. \tag{40}$$

$$q_{rxn,3} = 4.909 \times 10^{13} \text{ mol/(m}^3 \cdot \text{s)}. \tag{41}$$

Completed Table

Table 1: Calculated values for the three gas compositions.

Case	**1**	**2**	**3**
X_{CH_4}	0.147	0.147	0.005
X_{CO_2}	0.069	0.069	0.339
X_{H_2O}	0.460	0.579	0.396
X_{O_2}	0.324	0.206	0.260
$\Delta\bar{g}_{rxn}$ (kJ/mol)	-800.86	-791.77	-766.85
q_{rxn} (mol/(m$^3 \cdot$ s))	2.590×10^{15}	1.045×10^{15}	4.909×10^{13}

Trend Between $\Delta\bar{g}_{rxn}$ and q_{rxn}

The results show that all three cases have negative values of $\Delta\bar{g}_{rxn}$ and positive values of q_{rxn}. This means the reaction is thermodynamically favored in the forward direction, and the net reaction rate is also in the forward direction.

From Case 1 to Case 3, $\Delta\bar{g}_{rxn}$ becomes less negative,

$$-800.86 \rightarrow -791.77 \rightarrow -766.85 \text{ kJ/mol}, \tag{42}$$

and the rate of progress also decreases,

$$2.590 \times 10^{15} \rightarrow 1.045 \times 10^{15} \rightarrow 4.909 \times 10^{13} \text{ mol/(m}^3 \cdot \text{s)}. \tag{43}$$

Therefore, the common trend is that a larger negative Gibbs free energy corresponds to a larger positive reaction rate. As the system moves closer to equilibrium, $\Delta\bar{g}_{rxn}$ approaches zero and q_{rxn} also approaches zero.

At equilibrium, the thermodynamic driving force is zero and the forward and reverse reaction rates are equal. Therefore,

$$\Delta\bar{g}_{rxn} = 0 \tag{44}$$

and

$$q_{rxn} = 0. \tag{45}$$

Part II: Electrochemical Reaction

The same methane oxidation reaction is written as an electrochemical reaction in a single-chamber SOFC:

$$CH_4(an) + 2O_2(ca) + 4e^-_{ca} \rightarrow 2H_2O(an) + CO_2(an) + 4e^-_{an}. \tag{46}$$

The given electric potentials are

$$\phi_{an} = -0.8 \text{ V} \tag{47}$$

and

$$\phi_{ca} = 0.0 \text{ V}. \tag{48}$$

The problem defines the cell voltage as

$$V_{cell} = \phi_{ca} - \phi_{an}. \tag{49}$$

Therefore, for the given values,

$$V_{cell} = 0.0 - (-0.8) = 0.8 \text{ V}. \tag{50}$$

Net Electrochemical Driving Force

In Part I, only the chemical Gibbs free energy was considered. In this part, the electric potential also contributes to the total driving force because electrons are being moved from the cathode to the anode side.

For this reaction, four electrons are transferred. Therefore,

$$n = 4. \tag{51}$$

The net electrochemical driving force is written as

$$\Delta\tilde{\mu}_{rxn} = \Delta\bar{g}_{rxn} + nF(\phi_{ca} - \phi_{an}). \tag{52}$$

Substituting $n = 4$,

$$\Delta\tilde{\mu}_{rxn} = \Delta\bar{g}_{rxn} + 4F(\phi_{ca} - \phi_{an}). \tag{53}$$

Using

$$F = 96485 \text{ C/mol}, \tag{54}$$

the electrical contribution is

$$4F(\phi_{ca} - \phi_{an}) = 4(96485)(0.8) = 308.75 \text{ kJ/mol}. \tag{55}$$

So, for each case,

$$\Delta\tilde{\mu}_{rxn} = \Delta\bar{g}_{rxn} + 308.75. \tag{56}$$

Case 1

From Part I,

$$\Delta\bar{g}_{rxn,1} = -800.86 \text{ kJ/mol}. \tag{57}$$

Therefore,

$$\Delta\tilde{\mu}_{rxn,1} = -800.86 + 308.75 = -492.11 \text{ kJ/mol}. \tag{58}$$

Case 2

From Part I,

$$\Delta\bar{g}_{rxn,2} = -791.77 \text{ kJ/mol}. \tag{59}$$

Therefore,

$$\Delta\tilde{\mu}_{rxn,2} = -791.77 + 308.75 = -483.02 \text{ kJ/mol}. \tag{60}$$

Case 3

From Part I,

$$\Delta\bar{g}_{rxn,3} = -766.85 \text{ kJ/mol}. \tag{61}$$

Therefore,

$$\Delta\tilde{\mu}_{rxn,3} = -766.85 + 308.75 = -458.10 \text{ kJ/mol}. \tag{62}$$

Cell Voltage for Zero Reaction Rate

The reaction rate becomes zero when the electrochemical reaction reaches equilibrium. At that point, the net electrochemical driving force is zero:

$$\Delta\tilde{\mu}_{rxn} = 0. \tag{63}$$

Using

$$\Delta\tilde{\mu}_{rxn} = \Delta\bar{g}_{rxn} + 4FV_{cell}, \tag{64}$$

the equilibrium condition becomes

$$0 = \Delta\bar{g}_{rxn} + 4FV_{cell}. \tag{65}$$

Solving for the required cell voltage gives

$$V_{cell} = \frac{-\Delta\bar{g}_{rxn}}{4F}. \tag{66}$$

Since the Gibbs free energy values are in kJ/mol, converted to J/mol before calculating the voltage:

$$V_{cell} = \frac{-\Delta\bar{g}_{rxn}(1000)}{4F}. \tag{67}$$

Case 1

$$V_{cell,1} = \frac{-(-800.86)(1000)}{4(96485)} = 2.075 \text{ V}. \tag{68}$$

Case 2

$$V_{cell,2} = \frac{-(-791.77)(1000)}{4(96485)} = 2.052 \text{ V}. \tag{69}$$

Case 3

$$V_{cell,3} = \frac{-(-766.85)(1000)}{4(96485)} = 1.987 \text{ V}. \tag{70}$$

Completed Table

Table 2: Electrochemical driving force and equilibrium cell voltage for the three cases.

Case	**1**	**2**	**3**
$\Delta\bar{g}_{rxn}$ (kJ/mol)	-800.86	-791.77	-766.85
$\Delta\tilde{\mu}_{rxn}$ (kJ/mol)	-492.11	-483.02	-458.10
V_{cell} for $q_{rxn} = 0$ (V)	2.075	2.052	1.987

Observation

Compared with Part I, the electrochemical driving force is less negative because the electric potential adds a positive electrical work term. Even after including this electrical contribution, all three values of $\Delta\tilde{\mu}_{rxn}$ are still negative. This means the reaction is still favorable in the forward direction for the given anode and cathode potentials.

The calculated cell voltages are the voltages needed to make the net electrochemical driving force equal to zero. At those voltages, the forward and reverse reaction rates balance each other, so the net reaction rate of progress becomes zero.

Part III: Cantera Implementation

Before calculating the current, the same current sign convention was used as the Butler–Volmer expression given in the problem. With this convention, the Cantera-based current is calculated from the net rate of progress as

$$i = -nFq_{net}. \tag{71}$$

Part III(a): Current Density vs. Overpotential

For this part, the TPB charge-transfer reaction from the `sofc.yaml` file was used:

$$H(m) + O''(ox) \rightarrow OH(m) + (m) + 2e^-. \tag{72}$$

This reaction is listed as `edge-f2`. Since two electrons are transferred,

$$n = 2. \tag{73}$$

The overpotential was applied using

$$\Delta\phi = \phi_{Ni} - \phi_{YSZ} = \Delta\phi_{eq} + \eta, \tag{74}$$

where $\Delta\phi_{eq}$ was found by setting the Cantera net rate of progress equal to zero:

$$q_{net} = 0. \tag{75}$$

For each value of η, Cantera was used to calculate the net rate of progress. The current was then plotted as a function of overpotential, as shown in Figure 1.

Part III(b): Butler–Volmer Overlay

The Butler–Volmer current was calculated using the expression given in the problem:

$$i_{BV} = i^\circ \left[\exp\left(-\frac{\beta nF\eta}{RT}\right) - \exp\left(\frac{(1-\beta)nF\eta}{RT}\right)\right]. \tag{76}$$

For positive overpotential, the second exponential term is larger than the first term, so the Butler–Volmer current is negative. This is why the Cantera-based current was written as $i = -nFq_{net}$, so that both the Cantera calculation and the Butler–Volmer calculation use the same sign convention.

The exchange current density was evaluated from

$$i^\circ = nFk_f^{1-\beta}k_r^{\beta}\prod_i C_i^{(1-\beta)\nu_{i,f}}\prod_i C_i^{\beta\nu_{i,r}}. \tag{77}$$

The values of k_f, k_r, species concentrations, and stoichiometric coefficients were taken from Cantera. The Butler–Volmer curve was then plotted over the Cantera-based current calculation, as shown in Figure 2.

The close overlap confirms that the Butler–Volmer expression is consistent with the elementary charge-transfer rate used in Cantera.

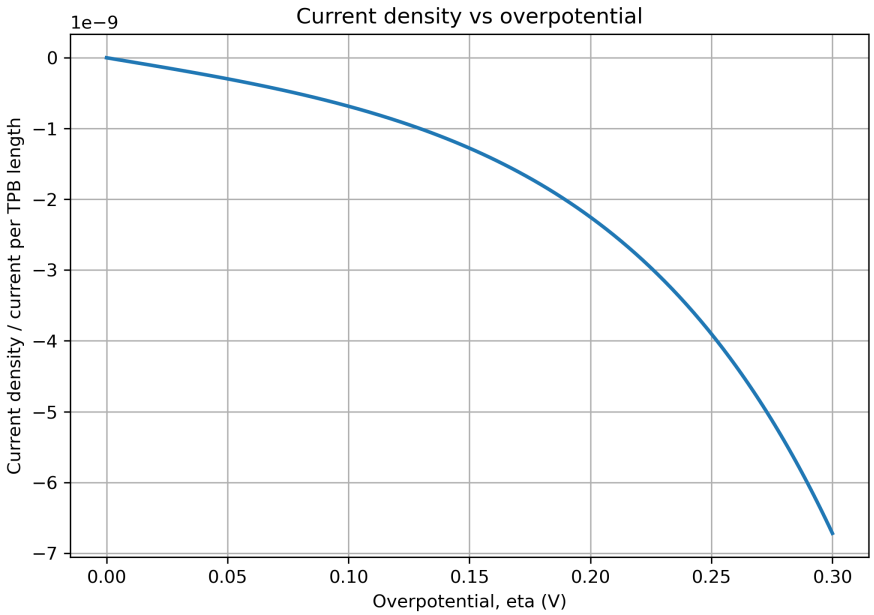

Figure 1: Current density as a function of overpotential.

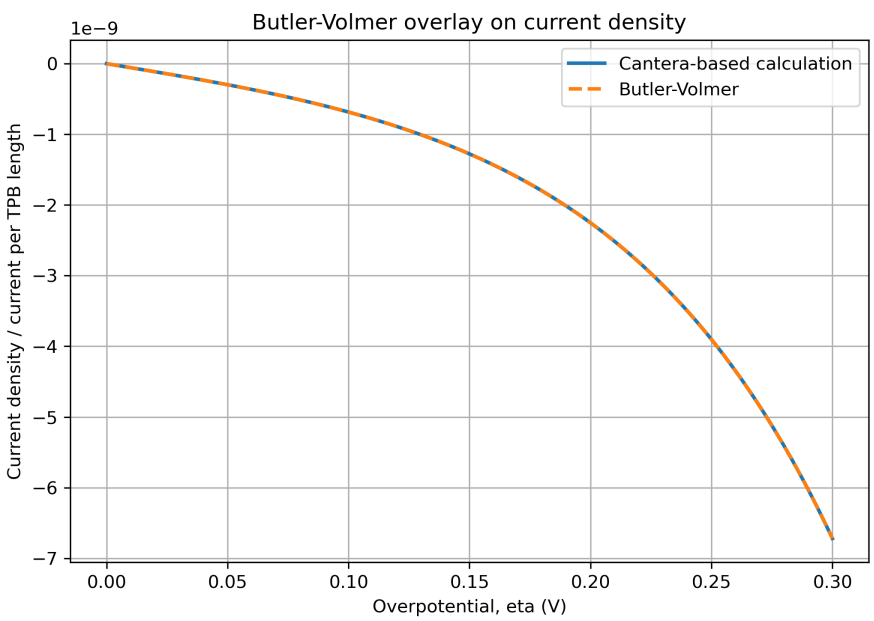

Figure 2: Overlay of the Cantera-based current calculation and Butler–Volmer current.

Part III(c): Constant i° Assumption

From the expression for i°, the exchange current density depends on the rate constants and surface species concentrations:

$$i^\circ = nFk_f^{1-\beta}k_r^{\beta}\prod_i C_i^{(1-\beta)\nu_{i,f}}\prod_i C_i^{\beta\nu_{i,r}}. \tag{78}$$

Therefore, i° is not generally constant. Assuming a constant value is only reasonable near a fixed operating point where temperature, gas composition, and surface coverages remain nearly unchanged.

This assumption can give significant errors when the operating conditions change strongly. For example, changes in temperature affect k_f and k_r, while changes in gas composition, oxygen activity, or surface coverage affect the species concentrations C_i. Since all of these quantities appear directly in the expression for i°, the exchange current density should be recalculated under changing operating conditions instead of treated as a constant.

```
In [24]: %runfile C:/Users/m10955083/Desktop/GIT/EChem/Assign3/sofc_part3_cantera.py --wdir
Selected reaction:
H(m) + O''(ox) <=> (m) + OH(m) + 2 electron
Reactants: {'H(m)': 1.0, "O''(ox)": 1.0}
Products : {'(m)': 1.0, 'OH(m)': 1.0, 'electron': 2.0}

Equilibrium result
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Delta_phi_eq = 0.37268897 V

Part III(a,b) summary
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Reaction used                        = H(m) + O''(ox) <=> (m) + OH(m) + 2 electron
Temperature, T                       = 1073.15 K
Number of electrons, n               = 2
Charge-transfer coefficient, beta    = 0.500
Delta_phi_eq                         = 0.37268897 V
Exchange current density, i0         = 2.625045e-10
Current density at eta = 0 V         = -8.960566e-18
BV current at eta = 0 V              = 0.000000e+00
Current density at eta = 0.3 V       = -6.719699e-09
BV current at eta = 0.3 V            = -6.719624e-09
```

Figure 3: values